

$\phi_B^{(11)} \rightarrow$

$$\begin{aligned} & \frac{1}{16\pi^2} \left(-\frac{1}{36} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_d^{\text{p}}] + \frac{1}{36} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_d^{\text{p}}] - \frac{1}{4} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_e^{\text{p}}] + \frac{1}{4} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_e^{\text{p}}] + \right. \\ & \frac{1}{16} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_l^{\text{p}}] - \frac{1}{16} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_l^{\text{p}}] - \frac{1}{144} \mathbf{g}_1^2 (6 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} + \sum_{\mathbf{p}} \mathbf{g}_1^2) \text{LF}_{3,0}[\mathbf{m}_q^{\text{p}}] + \\ & \frac{1}{144} (6 \mathbf{g}_1^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} + \sum_{\mathbf{p}} \mathbf{g}_1^4) \text{LF}_{4,-1}[\mathbf{m}_q^{\text{p}}] + \frac{2}{9} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_u^{\text{p}}] - \frac{2}{9} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_u^{\text{p}}] - \\ & \frac{2}{3} \mathbf{g}_1^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{3,0}[\mathbf{m}_u^{\text{r}}] + \frac{2}{3} \mathbf{g}_1^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{4,-1}[\mathbf{m}_u^{\text{r}}] - \frac{1}{6} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{2,2,0}[\mathbf{m}_d^{\text{r}}, \mathbf{m}_q^{\text{p}}] - \\ & \frac{1}{6} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_d^{\text{r}}, \mathbf{m}_q^{\text{p}}] + \frac{1}{6} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_d^{\text{r}}, \mathbf{m}_q^{\text{p}}] + \\ & \frac{1}{6} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_d^{\text{r}}, \mathbf{m}_q^{\text{p}}] - \frac{1}{2} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{2,2,0}[\mathbf{m}_e^{\text{r}}, \mathbf{m}_l^{\text{p}}] - \\ & \frac{1}{2} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_e^{\text{r}}, \mathbf{m}_l^{\text{p}}] + \frac{1}{2} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_e^{\text{r}}, \mathbf{m}_l^{\text{p}}] + \\ & \frac{1}{2} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_e^{\text{r}}, \mathbf{m}_l^{\text{p}}] + \frac{1}{2} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_l^{\text{p}}, \mathbf{m}_e^{\text{r}}] - \\ & \frac{1}{4} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_l^{\text{p}}, \mathbf{m}_e^{\text{r}}] + \frac{1}{4} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{5,1,-2}[\mathbf{m}_l^{\text{p}}, \mathbf{m}_e^{\text{r}}] + \\ & \frac{1}{3} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{r}}] + \frac{1}{6} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{r}}] - \\ & \frac{13}{12} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{r}}] + \frac{3}{4} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{5,1,-2}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{r}}] - \\ & \frac{1}{24} \mathbf{g}_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{2,2,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] - \frac{1}{24} \mathbf{g}_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] + \\ & \frac{1}{24} \mathbf{g}_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] + \frac{1}{24} \mathbf{g}_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] - \\ & \frac{7}{24} \mathbf{g}_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] + \frac{1}{24} \mathbf{g}_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] - \\ & \frac{11}{24} \mathbf{g}_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] + \frac{3}{4} \mathbf{g}_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{5,1,-2}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] - \\ & \frac{1}{8} \mathbf{g}_1^4 \text{LF}_{3,1,-1}[\tilde{\mu}, \mathbf{m}_1] + \frac{3}{8} \mathbf{g}_1^4 \text{LF}_{4,1,-2}[\tilde{\mu}, \mathbf{m}_1] - \frac{1}{4} \mathbf{g}_1^4 \text{LF}_{5,1,-3}[\tilde{\mu}, \mathbf{m}_1] - \\ & \left. \frac{3}{8} \mathbf{g}_1^2 \mathbf{g}_2^2 \text{LF}_{3,1,-1}[\tilde{\mu}, \mathbf{m}_2] + \frac{9}{8} \mathbf{g}_1^2 \mathbf{g}_2^2 \text{LF}_{4,1,-2}[\tilde{\mu}, \mathbf{m}_2] - \frac{3}{4} \mathbf{g}_1^2 \mathbf{g}_2^2 \text{LF}_{5,1,-3}[\tilde{\mu}, \mathbf{m}_2] \right) \end{aligned}$$